22AIE205 – Introduction to Python

Lab Activity 3

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Electricity Bill Generator (Slab-Based, Multiple Households)

The Electricity Board wants to calculate electricity bills for multiple households.

Store the electricity tariff slabs in a tuple.

Units rate (₹/unit),

First 100 ₹1.50,

Next 100 ₹2.50,

Next 300 ₹4.00,

Above 500 ₹6.00,

The program should:

i) Ask the user for the number of households.

ii) Store the household names in a list.

iii) Store the units consumed by each household in another list.

iv) Calculate the electricity bill according to the slab rates.

Display the details in a neat table.

If the PAN number starts with 'A', add an additional charge of ₹8 × Units Consumed.

CODE:

bill2.py > bill_calculate

```python
n = int(input("Enter no.of household names: "))
rates = (1.5, 2.5, 4.0, 6.0)
household_names = []
household_units = []
bill_total = []
pan_nums = []
extra_charge = []
bill_amount = []
print()
for i in range(0,n):
    print(f"Enter household name {i+1}: ")
    name =      input("Household Name: ")
    pan =       input("PAN Number    : ")
    units = int(input("Units Consumed:"))
    household_names.insert(i,name)
    pan_nums.insert(i,pan)
    household_units.insert(i,units)
    print()

def  bill_calculate(units,pan_num,i):
    extra_charge.insert(i,0.0)
    bill = 0
    if units<=100:
        bill = units*rates[0]
    if units>100 and units<=200:
        bill = 100*rates[0]+(units-100)*rates[1]
    if units>200 and units<=500:
        bill = 100*rates[0]+(100)*rates[1]+(units-200)*rates[2]
    if units>500 and units<=1000:
        bill = 100*rates[0]+(100)*rates[1]+300*rates[2]+(units-500)*rates[3]
    bill_amount.insert(i,bill)
    if pan_num[0]=='A':
        extra = units*8
        extra_charge.insert(i,extra)
        bill = bill+extra
    return bill

for i in range(0,n):
    calculated = bill_calculate(household_units[i],pan_nums[i],i)
    bill_total.insert(i,calculated)

total_revenue = sum(bill_total)
avg_bill = total_revenue/n
min_index = 0
max_index = 0
for i in range(1,n):
    if bill_total[i]>bill_total[max_index]:
        max_index = i
    if bill_total[i]<bill_total[min_index]:
        min_index = i

print(f"-------------------------------------------------------------------------------------------")
print(f"Household       PAN Number      Units       Bill(₹)     Extra Charge(₹)     Total Bill (₹)")
print(f"-------------------------------------------------------------------------------------------")
for i in range(0,n):
    print(f"{household_names[i]}          {pan_nums[i]}          {household_units[i]}          {bill_amount[i]}          {extra_charge[i]}")
print(f"-------------------------------------------------------------------------------------------")

print(f"Highest Bill    : {household_names[max_index]} (₹{bill_total[max_index]})")
print(f"Lowest Bill     : {household_names[min_index]} (₹{bill_total[min_index]})")
print(f"Total Revenue   : ₹{total_revenue}")
print(f"Average Bill    : ₹{avg_bill}")
```

OUTPUT:

```
PS D:\SEMESTERs\SEMESTER3\22AIE205-Python\090726Lab\090726Lab> python -u "d:\SEMESTERs\SEMESTER3\22AIE205-Python\090726Lab\090726Lab\bill2.py"
Enter no.of household names: 4

Enter household name 1:
Household Name: Ravi
PAN Number    : ABCTY1234D
Units Consumed:80

Enter household name 2:
Household Name: Kumar
PAN Number    : PQRST5678F
Units Consumed:250

Enter household name 3:
Household Name: Priya
PAN Number    : AXYPL9876M
Units Consumed:620

Enter household name 4:
Household Name: Arjun
PAN Number    : BHGTR4567K
Units Consumed:150

-----------------------------------------------------------------------------------------
Household       PAN Number      Units       Bill(₹)     Extra Charge(₹)     Total Bill (₹)
-----------------------------------------------------------------------------------------
Ravi          ABCTY1234D          80          120.0          640           760.0
Kumar          PQRST5678F          250          600.0          0.0           600.0
Priya          AXYPL9876M          620          2320.0          4960          7280.0
Arjun          BHGTR4567K          150          275.0          0.0           275.0
-----------------------------------------------------------------------------------------
Highest Bill    : Priya (₹7280.0)
Lowest Bill     : Arjun (₹275.0)
Total Revenue   : ₹8915.0
Average Bill    : ₹2228.75
```